Basics of Machine Learning

Machine learning (ML) is a subset of artificial intelligence where computers learn patterns from data to make decisions or predictions without explicit programming. It is transforming many industries by automating tasks like image recognition, language translation, and recommendation systems.

This presentation will guide beginners through ML fundamentals, different learning types, core libraries, and practical steps to build models using Python. Whether you want to enhance your data skills or start your ML journey, this overview sets a solid foundation.

By Swarnima Mitra

Introduction of Machine Learning

•We are in the *"age of data"*—huge volumes of data are generated daily.

•The challenge is not data collection, but making sense of it.

•Machine Learning (ML) is a key technology to extract patterns and insights from large datasets.

•ML is a subset of Artificial Intelligence (AI) that enables computers to learn from data.

•It uses well-defined algorithms to allow systems to improve performance based on experience.

•ML allows systems to make predictions or decisions without being explicitly programmed.

•It is widely used in applications like recommendation systems, fraud detection, speech recognition, etc

By Swarnima Mitra

Why Do We Need Machine Learning

- Humans are intelligent, but limited in handling massive data with speed and accuracy.
- Machine Learning allows machines to **think, analyze, and act** like humans on complex tasks.
- ML helps solve **data-driven problems** that are otherwise time-consuming or impractical for humans
- Enables automation of tasks that require decision-making based on patterns and data.
- Increases **efficiency and scalability** in solving problems.
- Useful in scenarios where manual decision-making is not feasible (e.g., real-time traffic navigation).
- Reduces **human error** in tasks such as medical diagnostics, financial forecasting, and cybersecurity.
- Allows systems to continuously **learn and improve** without needing to be reprogrammed.
- Essential for making **intelligent applications** like smart assistants (e.g., Siri, Alexa) and self-driving cars.

Common Algorithms

- Linear Regression
- Polynomial Regression
- Decision Tree Regression

Factors Driving the Need for Machine Learning

•**Lack of Human Expertise**:
In critical or unfamiliar environments (e.g., deep space, deep ocean, or disaster zones), ML can operate where humans cannot. *Example*: Autonomous underwater vehicles exploring the deep sea where divers cannot go

•**Safety Concerns**:
ML can take over tasks that are dangerous for humans, such as bomb disposal robots or chemical leak detection. *Example*: Robots with ML detecting and diffusing bombs to prevent human injury.

•**Dynamic Environments**:
ML adapts to rapidly changing data (e.g., stock markets, traffic patterns), where hard-coded programs would fail. *Example*: Real-time stock market trading algorithms adapting to rapid price changes.

•**Infrastructure Limitations**:
In remote or underdeveloped areas, human presence may not be feasible, but ML-powered systems can still operate. *Example*: Drones with ML delivering medical supplies in remote villages with no roads.

•**Accuracy Demands**:
ML can deliver high precision in tasks such as image classification, disease prediction, and quality control in manufacturing. *Example*: ML used in radiology to detect tumors in medical images with high precision.

Factors Driving the Need for Machine Learning

•**Volume, Variety & Velocity of Data**:
Traditional methods struggle with big data, but ML thrives by learning from and adapting to massive, diverse data in real-time. *Example*: Social media platforms using ML to analyze millions of posts in real time for trends and content moderation.

•**Cost and Time Efficiency**:
Automating decision-making processes reduces labor costs and speeds up analysis, especially in sectors like finance, healthcare, and e-commerce. *Example*: Chatbots using ML to handle thousands of customer queries instantly without human agents. *Example*: Chatbots using ML to handle thousands of customer queries instantly without human agents.

•**Personalization at Scale**:
ML enables services like personalized recommendations (e.g., Netflix, Amazon), tailored marketing, and adaptive learning. *Example*: Netflix recommending movies based on individual user viewing patterns.

•**Complex Problem Solving**:
ML models are used in fields like genomics, climate science, and logistics to find patterns and optimize outcomes beyond human capacity. *Example*: ML models predicting protein folding structures in bioinformatics (e.g., DeepMind's AlphaFold).

Machine Learning Model

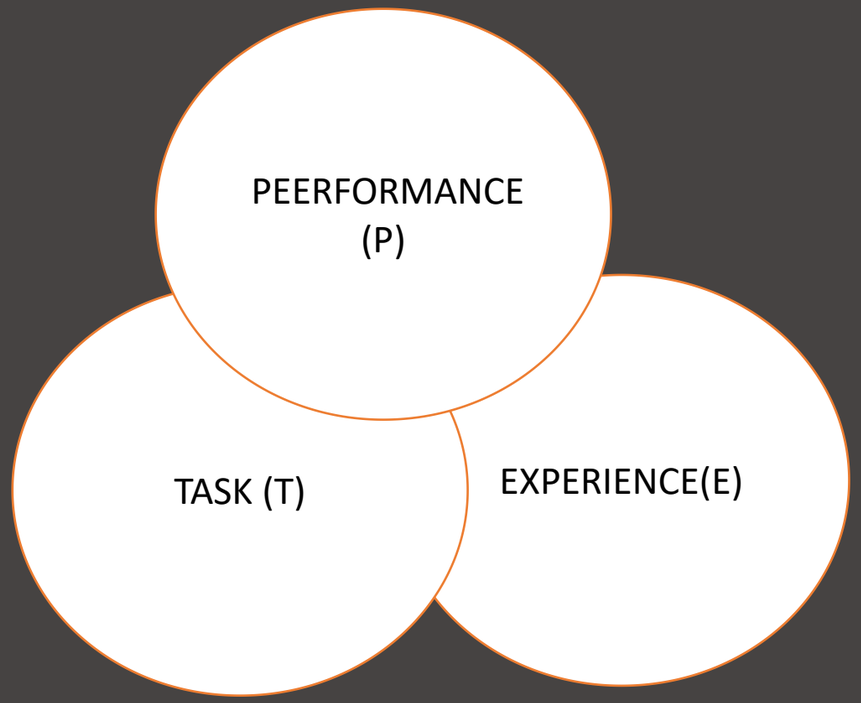
•A computer program is said to learn from experience (E) on tasks (T) and improve performance (P) over time.

•**Task (T)** refers to the specific problem like predicting sales of product and ML-based tasks are classification, regression, clustering, etc.

•**Experience (E)** is the data-driven knowledge that helps algorithms learn patterns and make decisions. Supervised, Unsupervised or Reinforcement learning are se ways to learn or grain experience. E used to solve T.

•**Performance (P)** measures how well the algorithm performs on task T using experience E (using metrics like accuracy, precision, recall, F1 score, sensitivity, etc.).

•Learning types include **supervised**, **unsupervised**, and **reinforcement learning**, which rely on various forms of feedback and data.



Challenges in Machine Learning

- Lack of **good-quality data** hinders effective ML model training and performance.
- Data collection and **preprocessing is time-consuming**, involving extensive feature extraction and cleaning.
- **Limited expertise** in the field, as ML is still evolving and not widely mastered.
- Challenges in **formulating business problems** clearly to apply ML techniques effectively.
- Issues like **overfitting**, **underfitting**, and handling **missing or irrelevant data** affect model accuracy and reliability.

Applications in Machine Learning

- Machine Learning is a **rapidly growing area** with wide-ranging applications.
- It's used to **analyze and solve real-world problems** using data.
- Common domains include **emotion recognition** from texts, images, or speech.
- Widely used in **sentiment analysis** for reviews, feedback, and social media content.
- These applications help organizations understand **user behavior and preferences** for better decision-making.

- Weather Forecastig an Prediction
- Object Recognition
- Fraud detection and prevention
- Spam Filtering
- HR systems use ML algo to identify characteristics of effective employees & us this knowledge to select best suitable candidate from the set of all the applicats of a particular vacant post

Understanding Supervised Learning

What is Supervised Learning?

This type of ML uses labeled data where input-output pairs are known. The model learns to predict the output from input features.

Regression Tasks

Regression predicts continuous values, like house prices or temperatures.

Classification Tasks

Classification assigns inputs to discrete categories, like spam detection or medical diagnosis.

Deep Dive into Classification

What is Classification?

Classification involves predicting discrete labels for data points, useful in fields like image recognition or disease detection.

Popular Classification Algorithms

- Logistic Regression
- Support Vector Machines
- Random Forests

Unsupervised Learning: Clustering Basics

What is Unsupervised Learning?

This learning type finds hidden patterns in unlabeled data without predefined outputs.

Clustering Overview

Clustering groups similar data points together, revealing natural structures in data.

Common Clustering Algorithms

Examples include K-Means, Hierarchical Clustering, and DBSCAN.

Key Python Libraries for Machine Learning

Scikit-learn

A versatile library offering simple and efficient tools for data mining and analysis.

Pandas

Provides powerful data structures for data manipulation and cleaning.

NumPy

Fundamental package for numerical computing in Python, enabling fast operations on arrays.

Hands-on with Basic Machine Learning Models

Data Preparation

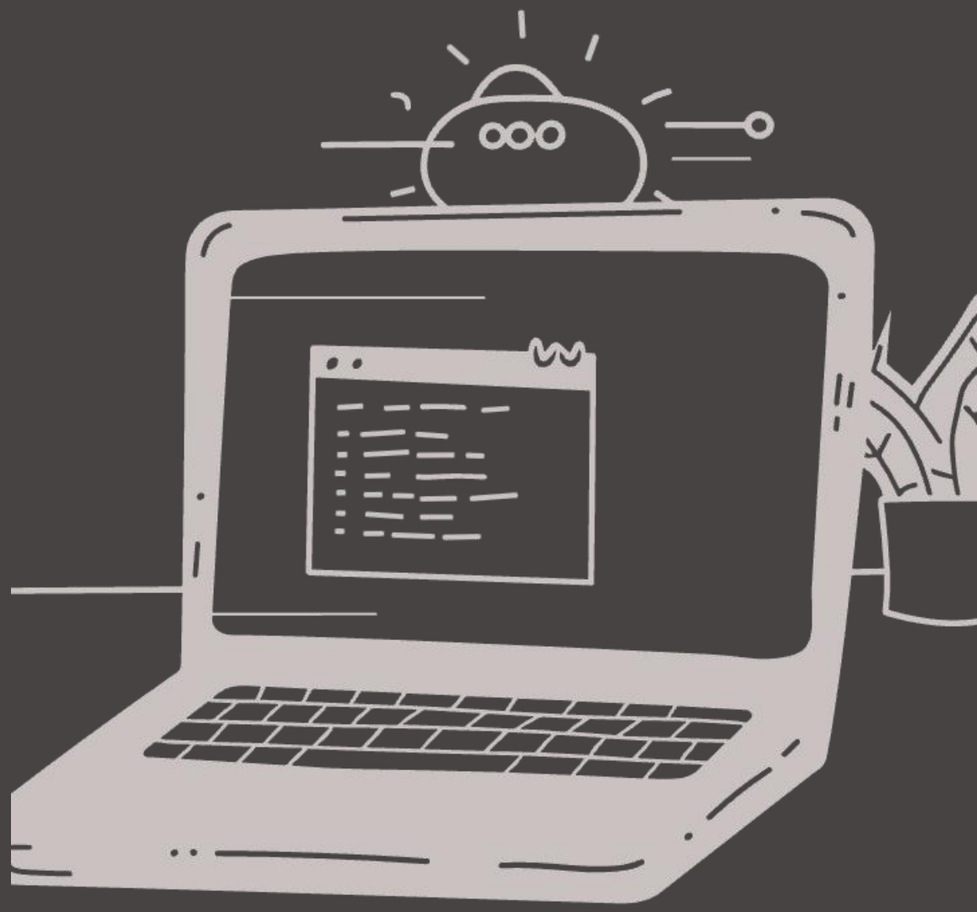
Collect and clean data, split into training and test sets.

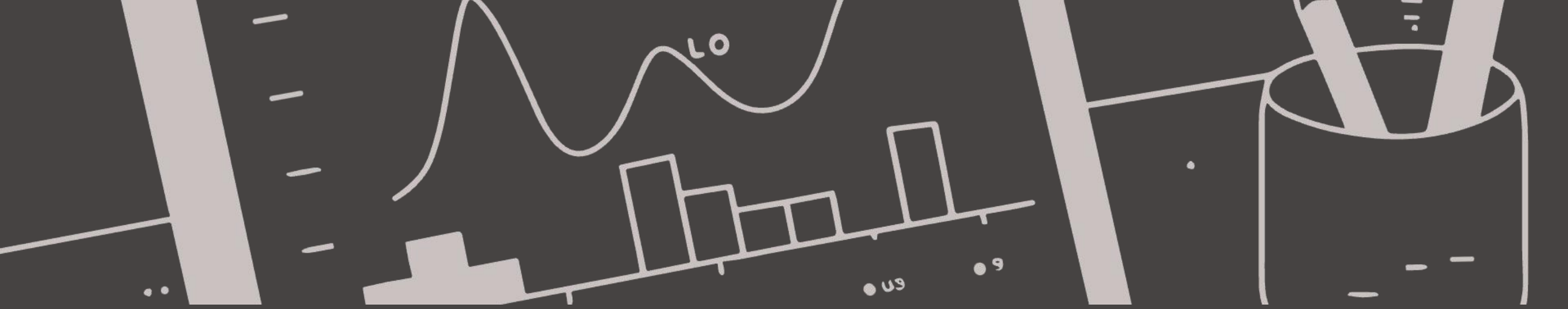
Model Building

Choose an algorithm, train the model using your training data.

Evaluation

Assess model accuracy on test data, and fine-tune if necessary.





Building a Regression Model with Scikit-learn

1 **Step 1: Import Libraries**

Load Scikit-learn, NumPy, and Pandas.

2 **Step 2: Load and Prepare Data**

Preprocess your dataset and split for training/testing.

3 **Step 3: Train Model**

Fit a linear regression model on training data.

4 **Step 4: Predict and Evaluate**

Use the model to predict on test data and measure error.

Building a Classification Model with Python

Gather and Prepare Data

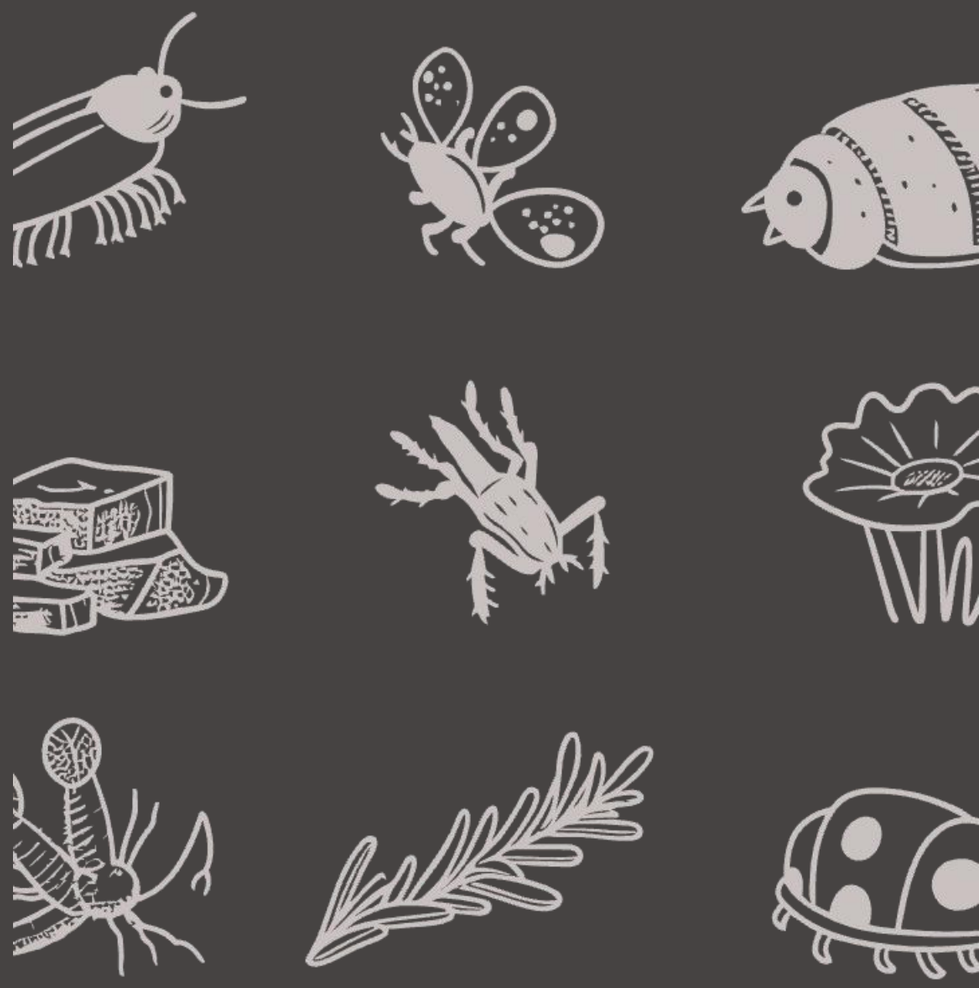
Obtain labeled data, clean it, and create training and test subsets.

Choose Algorithm

Select classifiers like Logistic Regression or Random Forest.

Train and Validate

Fit model to training set, then check accuracy on test set.



THANK YOU